INTRODUCTION

TUNING GUIDE

ABOUT THIS GUIDE

This is the long read behind Touge's Tune tab. The Tune tab tells you which setting to try for a symptom; this guide explains why it works in CarX Street, what happens when you go too far, and how the settings affect each other. Read it from start to finish if you want the whole picture, or open one chapter when a setting puzzles you.

Everything here is about how this game calculates the car, not how a real car works. Where the game behaves like a real car, the guide says so. Where it does not, the game wins.

WHERE THE KNOWLEDGE COMES FROM

- **The game's own physics code and data.** The formulas in the "How the game calculates it" sections come from the game's driving physics and its part data.
- **The native model.** The game's physics engine, run outside the game with a computer driver on a real route. It drives the same car with the same settings many times, so it can compare settings far more evenly than a person can.
- **Experience in game.** Setups driven and timed in the game itself, on touge routes, by a real driver.

> Markers: ● confirmed in game or in the game code, ◐ native model or inferred, not tested in game, ○ real-car theory, used only as background.

When the guide says "in our experience", the result comes from game tests or native runs on a handful of cars. Treat the numbers as an order of magnitude for your car, not a promise.

HOW THE GAME DIFFERS FROM A REAL CAR

A few things in the game physics change the usual tuning advice. They come up again in the chapters.

IN THE GAME	WHAT IT MEANS FOR TUNING
The roll centre is at ground level ●	All cornering load transfer goes through springs, bars and dampers. The bars do not reduce total transfer, they only decide which axle carries it.
Spring length is ride height and travel ●	Spring length is one of the biggest levers: it sets how much travel is left before the bump stop.
The damper knee is high, 0.1 m/s ●	The slow values do almost all of the handling work; the fast values act only on bumps, kerbs and gutters.
The game adds −3…+5° to the camber you set ●	Camber on screen is not the tyre's real camber; at rest it is about 2–3° more positive.
The diff lock is a share of driveline torque ●	The same lock locks much more in a low gear. Preload is the only lock with the clutch pressed.
Tyre grip grows with width to the power 0.41 ●, and the rim part sets the friction bonus ●	Width and the right rim part matter more than diameter.

HOW TO USE IT WITH TOUGE

1. **Build** the car on Touge's Build tab, or enter it from your screenshots.
2. **Start** from a preset such as touge weapon, then drive a few laps in the game.
3. **Find the phase** where the car does not do what you want: braking, turn-in, mid-corner, exit, straight, or bumps and gutters.
4. **Pick the symptom** in the Symptom view on the Tune tab. It shows the change to try first and second, and links to the chapter of this guide that explains it.
5. **Change one thing at a time** and drive the same corner in the same gear. Move the setting into the game with the To game view.

HOW TO READ A CHAPTER

Every chapter follows the same pattern:

1. **In short:** how big the lever is and the one thing to remember.
2. **How the game calculates it:** the formula or mechanism, with markers.
3. **When you raise:** what to expect, what happens when you go too far, and the experience behind it.
4. **Effect on other settings:** what else changes when you touch this one.
5. **Symptoms:** phase, symptom, what happens in the game, and what to change.
6. **Final tip:** how to approach it on a new car.
7. **Not yet confirmed:** what is still open. Read this before you trust a number.

CHAPTERS

1. The whole chassis: the biggest levers, setup order, how to test, and a symptom index.
2. Tyres: width, rim part, pressure, diameter and track.
3. Springs: spring length, stiffness and the front/rear balance.
4. Anti-roll bars: stiffness, roll and the front/rear split.
5. Dampers: slow and fast bump and rebound, and how they work together.
6. Camber and toe: tyre angle and wheel direction.
7. Differential: lock on acceleration, lock on deceleration and preload.
8. Body dimensions: wheelbase, track, centre of gravity and mass as a starting point.
9. Glossary: the guide's words and the game's setting names.

A WORD OF CAUTION

The fastest setup depends on the driver. When two options are within about half a second, keep the one that feels easier to control. Nothing in this guide replaces driving the change in the game.

CHAPTER 01

THE CHASSIS AS A WHOLE

IN SHORT

This guide pulls the topic guides together into one way of working. In this game the biggest levers are the bar split, spring length, tyre size and the diff's throttle lock: in our experience each has been worth seconds on a long touge run. Dampers, camber, toe and centre of gravity are finishing touches whose effect on time is often below the noise, but they change the feel.

Work from big to small, one change at a time, and solve a symptom with the lever that acts most on that phase. When two options are within about half a second, keep the one that feels easier to control.

> Markers: ● confirmed in game or in the game code, ◐ native simulation or inferred, not tested in game, ○ real-car theory.

Topic guides: Anti-roll bars · Springs · Tyres · Dampers · Camber and toe · Body dimensions · Differential

THE BIGGEST LEVERS

Lever size is the largest difference measured on a long touge run, in the native model or with the in-game timer. The figures come from different cars and starting points, so they show an order of magnitude, not a promise.

SETTING	LEVER SIZE	EXPERIENCE	ACTS MOST ON
Bar split	Large	◐ Up to about five seconds. ● Moving the split forward cured braking understeer and a nervous rear on a long car in game.	Balance through the whole corner

SETTING	LEVER SIZE	EXPERIENCE	ACTS MOST ON
Diff throttle lock	Large at the bottom, then flat	◐ 0 clearly slowest and an open diff far slower. Above that the results conflict (see Not yet confirmed). ● 0.24–0.30 has worked in game.	Exit
Spring length	Large	● A few millimetres more front length has been more than two seconds quicker in game.	Travel, bumps, throttle understeer
Tyre size	Large	◐ 285–325 mm vs 265 mm about two seconds.	Grip and gearing length
Final drive	Medium	◐ Short cars gained up to almost two seconds with a 5–10 % change.	Acceleration out of corners
Spring rate	Medium	◐ The extremes were 2–4 s slower than the middle of the range.	Body motion and bumps
Brake balance	Medium	◐ 40 % front was 0.5–0.7 s quicker on most cars. ● A front-biased balance has made corner entry safe in game.	Braking and turn-in
Dampers	Small for time, large for feel	◐ Nearly neutral in the native model. ● +20 % on the rear dampers has felt very good in game.	Speed of transitions
Track and offset	Small	◐ +0.10 m on both axles +1.4 % grip; on a full run often within noise.	Cornering grip
Camber and toe	Small	◐ Camber ≤ 3.5 % lateral grip, toe under a second.	Finishing
Centre of gravity, caster, Ackermann	Not measurable	◐ Lap within noise.	–

SETUP ORDER

The order goes from big to small, so that a later setting does not undo an earlier one. If you go back to an earlier step, check the steps after

it again.

1. **Tyres and body:** tyre size, rim part and offset inside the fender. The tyre sets grip and radius, so it is the base for everything else.
2. **Gearing:** only after the tyre, because only the total ratio (FD × tyre radius) matters ●.
3. **Spring length:** enough travel that the suspension does not hit the bump stop (touge weapon: front ×1.07, rear ×0.90 of the default).
4. **Springs:** rate by mass, then length again, because a stiffer spring raises the body ●.
5. **Bars:** split by wheelbase and total stiffness by roll. This is the main balance setting.
6. **Differential:** throttle lock first, then preload and decel lock for exit and entry.
7. **Brake balance:** rotation and stability on entry.
8. **Dampers:** slow bump and slow rebound for the speed of transitions. Fast values only for bumps and gutters.
9. **Camber and toe:** finishing.

The touge weapon preset in Touge gives starting values for steps 2–8 from the car's dimensions, and they have worked in game ●. The real refining then starts from symptoms.

HOW TO TEST

1. **One change at a time.** When you change two things you do not know which one made the difference. The exception is a pair that belongs together, such as a spring and its length for the same height.
2. **A big enough step.** One slider step splits into 20 ticks for gearing, springs, dampers, camber, toe and diff locks ●, so a gear ratio can be set to 0.4 %. The ratio the game shows is the physics ratio ●. A small step still disappears into noise. Useful steps: bar about 10 % of total stiffness or 2–4 points of split, spring length 4 mm, spring about 10 %, damper about 20 %, diff 0.05, brake balance 2 points, camber 0.5° and toe 0.1°.
3. **The same test spot.** Drive the same corner or section in the same gear. The full lap tells you the result, a single corner tells you what the change did.
4. **Know the noise.** In our experience runs of the same setup spread by about 0.8 s on a six-minute run ●. A difference under a second cannot be shown with one run.

5. **Note the feel as well as the time.** Write down what the car did under braking, at turn-in, mid-corner and on exit, not only the time. The Symptom view on Touge's Tune tab uses the same four phases.
6. **The whole lap decides.** A change is better only if it shortens the lap or makes the car clearly easier to control. Improving one measure, such as steady-circle grip, can slow the lap: rear toe-in raised circle grip 2–4 % but the lap got slower ◐.

On a controller the steering assist is always on ●. The steering assist's Off level does not give direct steering on a controller or keyboard. The game still uses the assist, with its tightest front slip limit (105 %). Direct steering is only available on a wheel. On a controller the Off/Gamepad profile is used, which has no extra keyboard stabilisation. In a native test the assist changed grip only a little (−1.2 % in a tight corner, 0 in a fast one) ◐, so it does not explain differences between the native model and the game. It may still affect the feel, because it keeps the front tyres near peak grip ◐.

FROM SYMPTOM TO SETTING

This index collects the symptom tables of the topic guides. Try the primary change first and only then the secondary one. The topic guides have more detail.

PHASE	SYMPTOM	PRIMARY CHANGE	SECONDARY
Braking into the corner	The car understeers on the brakes and has to be forced to turn	Move the bar split forward ●	Brake balance 2 points rearwards
Braking into the corner	The rear wanders or tries to overtake	Brake balance 2 points forward ●	Diff decel lock up, rear toe +0.1
Braking into the corner	The nose dives and bottoms out	Front spring length +4 mm	Front spring or front slow bump up
Turn-in	The car is lazy to turn	Front bar down or rear bar up	Front slow bump down, front camber −0.5°

PHASE	SYMPTOM	PRIMARY CHANGE	SECONDARY
Turn-in	The rear steps out right at turn-in	Rear bar down	Rear slow rebound down, brake balance forward
Mid-corner	Steady understeer in a long corner	Move the bar split rearwards	Tyre size or front track
Mid-corner, high speed	The rear lets go suddenly	Move the bar split forward by wheelbase ●	Wider rear track
Exit	The nose pushes on throttle in a fast corner	More front spring length ●	Front spring down ◐
Exit	The rear spins on throttle	Diff preload and throttle lock down (careful below 0.2)	Rear bar down
Exit	The inner wheel spins	Throttle lock up	Rear spring or rear length up, rear camber towards zero
Straight	The limiter comes before the braking point, or the car does not accelerate	Final drive	Tyre size (radius changes the gearing)
Bumps, kerbs and gutters	The car bounces or is thrown over the verge	More spring length	Fast bump down, spring down
Bumps, kerbs and gutters	The car keeps wallowing after a bump	Slow rebound up	Fast rebound up

If the same symptom shows in several phases, the cause is usually the bar split or the diff, not a single damper.

FINDING YOUR OWN FEEL

The fastest setup is not the same for everyone, because the driver drives it at the limit. First find a balance where the car does not surprise you, then tune the feel to your own style. When two options are within about 0.5 s, keep the one that is easier to control: with fewer mistakes it is quicker at the limit.

Feel adjusters:

YOU WANT	CHANGE	COST
More rotation on the brakes	Brake balance rearwards, diff decel lock down	The rear may step out under braking.
A more stable entry	Brake balance forward, decel lock up	The car understeers on the brakes.
A sharper response	Slow dampers up, bars stiffer with the same split	The car is more nervous over bumps.
Softer and more predictable	Slow dampers and springs down, more travel	Transitions get lazier.
An exit where the rear helps turn	Throttle lock and preload up	The rear steps out more easily on throttle.
A straight exit	Preload down, bar split forward	The car pushes more on throttle.

If you brake deep into the corner, use on/off throttle and drive over verges and gutters: brake balance becomes the main rotation adjuster, and a low diff decel lock keeps it from fighting the rotation on the brakes ●. Stability then comes from the bar split rather than the diff. Leave travel in the suspension for verges and gutters and keep fast damping moderate. With on/off throttle, a moderate preload and throttle lock slow the growth of a slide ◐.

FINAL TIP

Start a new car from the touge weapon preset, which has worked in game on ten cars ●. Drive a few laps and note the phase where the car does not do what you want. Fix one symptom at a time with the index's primary change, and once the car is balanced, move on to the feel.

Three things have repeated in every test:

1. **The bar split comes before anything else.** In our experience one split correction fixed the problems of nine cars at once.
2. **Travel comes before a low body.** More spring length has made the car seconds quicker even though the centre of gravity rose.
3. **The throttle lock should not be too low.** In the native model 0 was clearly slowest, and 0.24–0.30 has worked in game. Above that the

difference is mostly feel.

Simulation experience ◐ (native model, not tested in game): the build mattered most, then balance, and the diff last. Downforce, mass and tyre size moved the lap by 2–18 s, while chassis settings from a sensible start stayed under a second (the exception was a very soft rear bar, 3.7 s slower). More downforce on both axles was 2.8 s quicker; more only at the rear was 1.4 s slower (understeer in medium corners) and more only at the front 0.6 s quicker. 50 kg less mass was 2.1 s quicker. Balance changes saturate and do not add up. A final drive range of about 15 % stayed within 0.3 s.

NOT YET CONFIRMED

- The lever sizes come from different cars, different starting setups and partly the native model. They give the order, not how much your car will gain.
- The lower limit of the throttle lock conflicts: one model found below 0.25 slow, a hairpin test did not, another test found 0.10 slow, and a newer slide-capable driver saw 0.10–1.0 as noise over a lap with 0.15 quickest in the tightest hairpin. The 0.24–0.30 range that has worked in game is the safe area.
- Damper effect on time is small in the native model but large on feel. The native model does not model driver confidence.
- The feel adjusters are derived from the topic guides and real-car theory. They have not been tested together as a whole.
- The 0.8 s run-to-run spread is from one series on one car and track. It may differ elsewhere.

CHAPTER 02

TYRES

IN SHORT

The tyre is a big lever, and the most important choices are the width and the rim part. In our experience going from 265 mm to 285–325 mm has been worth about two seconds on a long touge run ◐. The same size often comes with two rim parts, and only the one with the lower base profile gives the full friction bonus of +0.08 ● game data. Diameter matters through wheel mass and ride height, and spacers add a little grip.

Rule: tyres must not stick out of the fenders, even though the game allows it.

> Markers: ● confirmed in game or in the game code, ◐ native simulation or inferred, not tested in game.

CHOOSING THE SIZE

How the game calculates it:

1. Tyre friction = 1 + the rim part's friction bonus (fric_add), and it scales the tyre's peak force directly ●. The game shows this as the grip value (for example 1.08).
2. The same size often has two rim parts. The one with the lower base profile gives +0.08, the higher one +0.03–0.05 ●. For example 265 R16: base profile 40 = +0.08, base profile 45 = +0.03.
3. The full +0.08 starts at 265 mm. 245–255 mm gets at most +0.05 and 225–235 mm +0.04 ●.
4. Width adds grip with a power of 0.41: grip factor = (width / 185)^0.41 × friction ● (route model, matches the native code).
5. Tyre grip peaks at 2.0 bar. 1.0 or 3.0 bar costs about 1–2.5 % grip ●.

WIDTH	HIGHEST FRICTION BONUS	GRIP FACTOR	DIFFERENCE TO 265
225 mm	+0.04	1.13	−10 %

WIDTH	HIGHEST FRICTION BONUS	GRIP FACTOR	DIFFERENCE TO 265
245 mm	+0.05	1.18	−6 %
265 mm	+0.08	1.25	0
285 mm	+0.08	1.29	+3 %
325 mm	+0.08	1.36	+9 %
345 mm	+0.08	1.39	+11 %

When you go wider:

CHANGE	EXPECT	TOO MUCH	EXPERIENCE
Wider tyre	More corner speed and braking grip, most in slow corners.	A heavier wheel slows acceleration, and the tyre may stick out of the fender.	◐ 265 → 285 was worth almost two seconds, 285 → 325 only about half a second more. ◐ In a tight corner the gain was about ten times larger than in a fast one.
Lower base profile in the same size	Friction +0.08 instead of +0.03–0.05.	—	● Rim part data. ◐ Worth close to a second on a full run.

By the grip factor the gain would keep growing evenly, but in the native model the width gain shrinks above 285 mm. The reason is still open ◐.

SMALL VS LARGE TYRE

How the game calculates it:

1. Diameter = rim × 25.4 + 2 × width × profile / 100 mm ●. The game's Tire profile −10…+10 is profile percentage points (285/35 → 285/25…285/45) ●.
2. Wheel mass comes from the size: m = 1000 · r² · width kg, and inertia I = 0.7 · m · r² ●. Under acceleration each wheel also weighs 0.7 × its own mass extra because it spins ●.
3. A larger tyre raises the body by the radius change: on a 285 mm tyre 2.85 mm per profile point ●.
4. In the native model only the total ratio (gear × final drive × radius) matters ●. A larger radius makes the gearing longer, which the final

drive can correct: FD_new = FD × r_new / r_old. The wheel rolls on radius r – load / vertical stiffness ●.

5. Profile also changes the tyre's vertical stiffness and its Pacejka defaults ●, but with ride height held constant the grip difference stayed within −0.7…+1.2 % ◐.

TYRE	RADIUS	WHEEL MASS
185/50 R14	0.27 m	13.5 kg
265/30 R16	0.28 m	21.2 kg
325/25 R17	0.30 m	28.7 kg
285/35 R18	0.33 m	30.7 kg

When you go larger (diameter up):

CHANGE	EXPECT	TOO MUCH	EXPERIENCE
Larger rim or profile	The body rises, the gearing gets longer and the wheel weighs more.	Weaker acceleration, a higher centre of gravity, and too long gears without an FD correction.	◐ Same radius, heavy wheel 30.7 vs 13.5 kg: 0–100 km/h +5 %. ◐ Three rim sizes with the same radius gave the same result.
Profile up (Tire profile +)	The body rises about 2.8 mm per step and the tyre flexes more.	A high body if the spring length is not shortened.	◐ Profile ±10 with ride height held constant: at most ±2 s on a ten-minute run.

A smaller and lighter wheel accelerates better, but the difference is small, so a heavy wheel is no reason to avoid a wide tyre ◐. Choose the size by grip, then correct the radius with the final drive and the height with the spring length.

SPACERS AND TRACK WIDTH

How the game calculates it:

1. Wheel offset (cm) = the body kit's fender part offset + the spacer value ●. The game rounds the white number in Part Setup to an integer; the exact value is only visible in the tyre component view ●.

2. The wheels sit at x = ∓ track / 2 ●, and track = the car's base track + offset (factor 1 is inferred ◐). Spacers do not change mass ●.
3. Tyre width does not move the wheel ●. A wider tyre does not widen the track; it fills the fender in both directions.
4. A wider track lowers the whole car's load transfer m · a · h / t ● and stiffens the axle in roll with the square (k · t²) ●. The widened axle takes a larger share of the roll moment, so track also works as a balance setting, like a bar.
5. The body kit's only physics stat is track width: it does not change drag or downforce ●.

When you widen (offset up):

CHANGE	EXPECT	TOO MUCH	EXPERIENCE
Both axles	A little more lateral grip and less roll.	The tyre sticks out of the fender (not allowed).	◐ +0.10 m +1.4 % grip, +0.26 m +3.2 %. ◐ On a full run the gain showed only with narrow tyres and stock bars; otherwise it was within noise.
Rear only	The rear gains grip and the car understeers slightly more.	The nose pushes.	◐ Rear +0.20 m: grip +2.7 %, front share of roll stiffness 0.555 → 0.520.
Front only	The nose gains a little grip.	The rear gets relatively lighter.	◐ Front +0.20 m: grip +0.5 %, front share → 0.587.

The widening gain shrinks with tyre width: +0.20 m gave +2.6 % on a 185 mm tyre, +1.9 % on 245 mm and +1.6 % on 285 mm ◐. With a wide tyre, use offset first of all to fit the tyre inside the fender.

EFFECT ON OTHER SETTINGS

Setup order: choose the tyre size and rim part first, then the offset so the tyre fits, then the final drive and spring length for the new radius.

SETTING	HOW IT LINKS TO THE TYRES IN THE GAME PHYSICS
Gearing	Only the total ratio matters ●, so a radius change is corrected with the final drive.

SETTING	HOW IT LINKS TO THE TYRES IN THE GAME PHYSICS
Spring length	A larger radius raises the body by the same amount ●. Shorten the spring if you want the same height.
Bars and springs	The tyre's vertical stiffness (p + 30 kPa) · (10 · width + 1) · r · 0.8 is in series with the suspension ●. A wide tyre is stiffer, and track stiffens roll with the square. Width and bar split overlapped by about a second in our experience ◐.
Tyre pressure	Grip peaks at 2.0 bar ●. Pressure lowers rolling resistance (∝ 1 / pressure) ●, so 3.0 bar is fastest on a straight (0–150 km/h about 7 % quicker than at 2.0 bar). Tyre type does not change the pressure effect, and wear does not cost grip before condition reaches 0 ●.
Mass	A wide and large wheel weighs more. Mass costs about 2 % grip per 100 kg ◐, but the extra wheel mass is small next to the width gain.
Camber	Camber costs 0.5 % longitudinal grip per degree and does not raise the lateral peak ●.
Kerbs and gutters	Leaning on the rising outer edge of a corner added +14…+55 % grip in the native model, and an inside gutter +26 % ◐. The effect is larger than the tyre size.

SYMPTOMS

A tyre choice changes the grip level of the whole car, so symptoms usually show at the end with the narrower tyre or the lower-friction rim part.

PHASE	SYMPTOM	WHAT HAPPENS IN THE GAME	CHANGE
Braking into the corner	Long braking distance and ABS working hard	The tyre's peak force is low	Check the rim part's friction (+0.08?) and the width.
Mid-corner	The whole car slides out evenly in slow corners	The grip level is low and neither end stands out	Wider tyres, 265 mm or more. Width helps most in tight corners ◐.
Mid-corner	The nose or the rear clearly gives up first	The axles differ in tyre width or friction	Even out the friction (same rim part type front and rear) and correct the rest with the bars.

PHASE	SYMPTOM	WHAT HAPPENS IN THE GAME	CHANGE
Exit, full throttle	The car accelerates lazily onto the straight	The total ratio got longer with the larger radius, or the wheels are heavy	Correct the final drive: FD × r_new / r_old.
Straight and top speed	Top speed changed after a tyre change	The total ratio changed	Correct the final drive.
Bumps, kerbs and gutters	The body hits the ground or the car sits too high	The radius changed the ride height	Correct with the spring length.

FINAL TIP

The widest tyre that fits inside the fender, with the lower-base-profile rim part (friction +0.08), gives the most grip. After choosing it, correct the final drive for the radius and the ride height with the spring length. Above 285 mm the native model shows only a small further gain.

Simulation experience ◐ (native model, not tested in game): pressures 2.5 and 3.0 bar were equally fast. 3.0 → 2.0 bar was 1.2 s slower and 1.5 bar 2.6 s slower, because rolling resistance outweighed the grip gain. Lowering the rear pressure cost more than lowering the front. 265/30 R16 was 1.9 s and 285/30 R16 1.1 s slower than 325/25 R17.

NOT YET CONFIRMED

- Why the width gain shrinks above 285 mm in the native model is open ◐.
- The track factor 1 (offset 10 cm = track +10 cm) is inferred, not read from the code ◐.
- Tyre wear lowers friction (up to 40–60 % depending on the tyre series), but the exact formula is unverified and is not modelled in Touge or in native runs.
- That width helps most in tight corners rests on a single native comparison.

CHAPTER 03

SPRINGS

IN SHORT

The biggest spring lever is length, because in this game it is the ride height. In our experience a few millimetres more front spring length has been worth seconds on a twisty road ●. Stiffness is a medium lever: clearly softer or stiffer than the car likes costs 2–4 s ◐. The front/rear ratio matters less than the bar split.

The game's springs have two special features compared with a real car. The spring rate is the wheel rate directly, with no motion ratio, and a stiffer spring lifts the body, because the length is the spring's free length.

> Markers: ● confirmed in game or in the game code, ◐ native simulation or inferred, not tested in game.

LENGTH

How the game calculates it:

1. Spring Length is the spring's free length in millimetres ●. At rest the spring is compressed by the load: sag = wheel load / spring rate ●.
2. The spring's top mount sits at the wheel, not at the length. The body and the centre of gravity therefore rise exactly 1 mm for every millimetre of length ●. In the game, length is a ride-height adjustment.
3. Length also sets the travel: the suspension hits the bump stop when the wheel load exceeds static + k · (length − sag) ●. On a bumpy road the front load can briefly reach about 3.6 × static, so travel without bottoming is about 3.6 × sag ◐.
4. Camber changes with length: camber = setting − 3° + 8° · length / stroke ●. A longer spring makes the resting camber more upright, but camber's grip effect is small ◐.

5. On a flat circle, ride height 70–120 mm did not change grip ◐. The difference comes from bumps, kerbs and whether the suspension hits the bump stop.

When you raise:

CHANGE	EXPECT	TOO MUCH	EXPERIENCE
Front length	The nose gets travel: less bottoming and less throttle understeer in fast corners.	A high nose and centre of gravity, more roll and lazier transitions.	● +4 on the front was worth over two seconds on a twisty road; the next +4 was even. ◐ Lowering both ends by 15 cost about 1 s.
Rear length	The rear gets travel and calms down over bumps.	A high rear moves weight forward under braking and makes the rear light.	◐ A large rear length change was within noise on the lap.
Length down (both)	A lower centre of gravity and less load transfer.	The suspension hits the bump stop and the car feels stiff.	◐ More travel cut bump-stop time by three quarters. The game's own tuning help warns about losing control on bumpy roads with too short springs.

STIFFNESS

How the game calculates it:

1. Spring force F = k · (length − current length) goes straight to the wheel ●. Game value × 1000 = wheel rate in N/m, with no motion ratio ●. Spring rates of different cars can therefore be compared directly against mass.
2. A stiffer spring sags less, so with the same length the body rises ●. When you stiffen, shorten the length by the same amount if you want to keep the height.
3. Springs do not change total load transfer (m · a · h / t sideways, m · a · h / L lengthways), because the roll centre is at ground level ●. They decide how much the body moves and how well the wheel follows bumps.
4. In roll the spring pair gives k · t² / 2 and a bar k · t² ●. The same number in a bar stiffens roll twice as much as in a spring.
5. Natural frequency f = √(k / m) / 2π, where m is the mass per wheel ◐. Frequency is the easiest way to carry a working stiffness over to

another car.
6. Helper is a progressive helper spring, not a driving aid ●. The slider limits come from the suspension part, and the game can limit more than the data says ●.

When you raise:

CHANGE	EXPECT	TOO MUCH	EXPERIENCE
Both springs up	Less body movement, a sharper response, and the body rises (see point 2).	The wheels do not follow bumps, and the car bounces on kerbs and gutters. Very stiff combinations were unstable in the simulation.	◐ Clearly stiffer than the car likes cost about 2 s. ● A moderately stiffer spring gave "attack and grip into corners" in game.
Both springs down	The wheels follow the road and the car calms down over bumps.	The body sinks and hits the bump stop; roll and pitch grow.	◐ A much softer spring cost up to 4 s.

FRONT/REAR RATIO

How the game calculates it:

1. The spring pair is part of axle roll stiffness ($k \cdot t^2 / 2$), so the spring ratio moves the load-transfer split in the same direction as the bars ●. A stiffer front spring moves the limit to the nose, a stiffer rear spring to the tail.
2. The effect is smaller than with the bars, because the spring pair counts half of the same number in a bar ●. In our experience changing springs did not cure a nervous rear; the bars did ◐.
3. Under braking and acceleration the longitudinal transfer $m \cdot a \cdot h / L$ is the same whatever the springs ●. The spring ratio decides how much the nose dives and the rear lifts, and with that the travel and camber.
4. Frequency describes the ratio better than the plain number: a front-heavy car needs a stiffer front spring for the same frequency ◐.

When you raise:

CHANGE	EXPECT	TOO MUCH	EXPERIENCE
Front spring (rear unchanged)	The nose dives less under braking and turn-in gets slightly sharper.	Throttle understeer in fast corners, and the nose bounces over bumps.	◐ A small effect on the lap; softer front reduced throttle understeer.
Rear spring (front unchanged)	The rear rotates the car and throttle understeer drops.	The rear gets light, the inside wheel spins and the rear bounces over bumps.	● A softer rear spring together with bar and diff changes was over a second faster in game. ◐ Clearly softer rear was slower.

Fix the balance with the bars first. Change the spring ratio only when the symptom is about bumps or about pitch under braking and throttle.

EFFECT ON OTHER SETTINGS

Order: length first (travel and height), then stiffness, and balance with the bars. Every spring change also moves the ride height, so check the length after a stiffness change.

SETTING	HOW IT LINKS TO THE SPRINGS IN THE GAME PHYSICS
Anti-roll bars	Springs and bars form the roll stiffness together ●. If you change springs, move the bars so the split stays.
Dampers	The right damping level depends on the spring. Too much rebound packs the suspension down onto the bump stop ◐. When you stiffen the spring, raise damping in roughly the same ratio.
Centre of gravity	Length raises the centre of gravity 1 mm per mm ●. More height adds load transfer, but on a flat circle grip did not change ◐.
Camber	Camber depends on spring length ●. On a rigid-axle car camber has no effect ●.
Tyre profile	Profile raises the body ●. A bigger tyre needs a shorter spring for the same height.
Differential	A soft or high rear rolls and unloads the inside wheel, so the throttle lock engages earlier ◐.
Bumps and gutters	Road bumps are small and frequent plus larger and long ●. A too stiff or too short spring cannot follow them, which shows as bouncing on kerbs.

SYMPTOMS

Change one thing at a time: length by 4, stiffness by about 10 %. Drive the same corner in the same gear.

PHASE	SYMPTOM	WHAT HAPPENS IN THE GAME	CHANGE
Braking into the corner	The nose dives and the car feels hard or pushes at the end of braking	The front suspension hits the bump stop when travel runs out	Lengthen the front by 4. If the nose still dives a lot, raise the front spring about 10 %.
Turn-in	The car is lazy to turn and rolls a lot	The body moves more than the tyre can follow	Raise the springs about 10 % or shorten the length if there is travel to spare.
Mid-corner	The car is nervous and bounces over bumps in the corner	Too stiff a spring; the wheel leaves the road	Lower the spring about 10 % and keep the height with length.
Exit, full throttle	The nose pushes on throttle in a fast corner	The nose rises and unloads; the front suspension does not move	Lengthen the front ● or soften the front spring ◐.
Exit, opening the throttle	The rear squats and the inside wheel spins	A soft or short rear sinks, and roll unloads the inside wheel	Raise the rear spring about 10 % or the rear length by 4.
Bumps, kerbs and gutters	The car bounces or is thrown over the verge	The suspension bottoms or the spring is too stiff to follow	Lengthen first. If the car still bounces, lower the spring.

FINAL TIP

Start with length: give the suspension enough travel not to hit the bump stop, and still keep the body low. Then change stiffness and correct the height with length. A slightly longer front and slightly shorter rear than stock has worked in game on several cars ●.

The spring value is the wheel rate, so a heavier car needs a stiffer spring for the same feel.

Simulation experience ◐ (native model, not tested in game): lower was slower: both lengths −15 cost about 1 s, and +15 on the front gained about 0.5 s, the same direction as in game. A stiffer rear spring gained a little and a softer one lost almost a second. A soft front spring with a stiff front bar was a poor combination (+1.7 s).

NOT YET CONFIRMED

- The flat simulation circle does not show bumps and gutters, which is exactly where spring and length are felt most. The game decides.
- Frequency and sag figures are calculated, not measured.

CHAPTER 04

ANTI-ROLL BARS

IN SHORT

In this game the anti-roll bars are the biggest chassis lever, and the front/rear split matters more than the total stiffness. In our experience the split alone can move a lap by several seconds ◐, and moving the split towards the front has cured a nervous rear in game ●. A bar only works when the left and right wheels move differently: in roll and over one-sided bumps.

> Markers: ● confirmed in game or in the game code, ◐ native simulation or inferred, not tested in game.

STIFFNESS AND ROLL

How the game calculates it:

1. The bar is a spring between the left and right wheel: F = k · (x_right − x_left) ●. When both wheels move the same amount (straight braking, acceleration, a flat bump), the bar does nothing.
2. Game value × 2000 = bar stiffness in N/m ●. One slider step doubles the value; the minimum is half of the stock value ●.
3. Axle roll stiffness = bar $k \cdot t^2$ + spring pair $k \cdot t^2/2$, and the tyre's vertical stiffness is in series with these ● (t = track width). The same bar is stiffer on a wider track, by the square.
4. The roll centre is at ground level ●. Total load transfer in a corner is therefore always m · a · h / t, and the bars do not reduce it. They only decide how much the body rolls and which axle carries the transfer.
5. Roll changes camber through spring travel ●, but camber's effect on grip is small ◐.

When you raise both by the same ratio (split stays):

CHANGE	EXPECT	TOO MUCH	EXPERIENCE
Both bars up	Less body roll and a quicker steering response. The limit hardly changes, because total transfer stays the same.	One-sided bumps and gutters throw the car; the inside wheel unloads and spins.	● A very stiff rear bar makes the car sensitive to bumps. ◐ Both very soft was slower and unstable.
Both bars down	More roll, and the car calms down over bumps.	Transitions (S-bends) get lazy and the car wallows.	◐ The native flat circle does not show bump sensitivity; the game decides.

FRONT/REAR SPLIT

How the game calculates it:

1. The front axle's share of roll stiffness (LLTD) equals its share of the cornering load transfer ●. LLTD 0.40 = the nose carries 40 % of the transfer and the rear 60 %.
2. The axle that carries more transfer loses grip relative to the other: its inside wheel unloads, and the tyre's lateral stiffness does not grow at the same rate on the outside wheel ● formula, ◐ interpretation. A stiffer front moves the limit to the nose, a stiffer rear to the tail.
3. The split changes the car's whole limit: in the simulator the same car ran 1.08–1.64 g by the bar split alone ◐. Where the limit sits depends on the car and its centre of gravity.
4. In our experience a longer wheelbase wants a more front-biased split than a short car ●. A split that works on a short roadster does not carry over to a longer car as such.
5. In normal driving (0.8 g) the steering angle is the same with every split ◐. The split is felt near the limit and in the braking and throttle transitions.

When you raise:

CHANGE	EXPECT	TOO MUCH	EXPERIENCE
Front bar	The nose carries more transfer. The rear calms down in fast corners and the limit comes from the nose. On a long car, braking and trail-brake understeer also drop.	The nose pushes in steady corners and on throttle in fast corners.	● Moving the split forward cured a nervous rear and braking understeer in game. ● A too-stiff front bar has cost almost two seconds.
Rear bar	The rear rotates the car: sharper turn-in and less throttle understeer. Faster on a short car.	The rear lets go "like a switch" in side load, a long car understeers on the brakes, and mid-corner throttle spins it.	● A soft front and a stiff rear has been fastest on short cars. ◐ Softening a stiff rear bar a lot cost several seconds.

The direction is the same as in a real car, but the limit is car-specific. Start from the car's wheelbase, not from a general "stiffer front" rule.

EFFECT ON OTHER SETTINGS

Set the split first, then the total stiffness. Springs and track also change roll stiffness, so check the split after changing them.

SETTING	HOW IT LINKS TO THE BARS IN THE GAME PHYSICS
Wheelbase	A longer car needs a more front-biased split ●.
Centre of gravity height	Load transfer is directly proportional to height ●, so the split matters more on a tall car.
Front weight	LLTD is compared with the front weight share. More front weight and a front-biased split move the same balance ◐.
Springs	The spring pair is part of roll stiffness, so a stiffer rear spring is also a stiffer rear ●. In our experience the bars cure balance better than the springs ◐.
Track and offset	Roll stiffness grows with the square of track width ●. A wider rear track takes more roll moment but changes wheel load less, and adds rear grip ◐.

SETTING	HOW IT LINKS TO THE BARS IN THE GAME PHYSICS
Tyres	The tyre's vertical stiffness is in series with the bar ●. A soft or low-pressure tyre softens the axle slightly.
Differential	A stiff rear unloads the inside wheel, which spins, and the throttle lock engages earlier ◐. A stiff rear bar needs a throttle lock of at least about 0.3 ◐.
Dampers	The bar decides how much load moves, the slow dampers how fast. A soft front rebound reduces throttle understeer in the same direction as a soft front bar ◐.

SYMPTOMS

First find the corner phase where the symptom appears. Change one bar at a time by about 10–15 % and drive the same corner in the same gear.

PHASE	SYMPTOM	WHAT HAPPENS IN THE GAME	CHANGE
Braking into the corner	The car will not turn on the brakes; trail braking needs force	On a long car a rear-biased split puts the limit at the wrong end ◐ ●	Raise the front bar and lower the rear.
Turn-in	The rear steps out right at turn-in	The rear carries too much transfer	Lower the rear bar one step (10–15 %).
Mid-corner	The nose pushes in a steady corner	The nose carries too much transfer	Lower the front bar ●.
Mid-corner	In a fast corner the rear lets go "like a switch"	The limit comes from the rear ◐	Raise the front bar or lower the rear.
Exit, full throttle	The nose pushes on throttle in a fast corner	The front is too stiff relative to the rear ◐	Lower the front bar. If it is already low, soften the front rebound.
Exit, opening the throttle	The rear spins mid-corner on throttle	A stiff rear unloads the inside wheel and the diff locks	First lower diff preload and throttle lock ◐, then the rear bar.

PHASE	SYMPTOM	WHAT HAPPENS IN THE GAME	CHANGE
Bumps, kerbs and gutters	A one-sided bump or gutter throws the rear	The bar passes one wheel's hit to the other	Lower the rear slow rebound, then the rear bar ◐.

FINAL TIP

Find the split first and the total stiffness after it. Start from a split that suits the car's wheelbase and refine by symptom, one bar at a time. A short car tolerates a stiff rear and gets faster with it; a long car needs front ≥ rear.

A working bar setup is only valid with that car's own springs and tyres, because springs and track are part of the same roll stiffness. If you change springs, move the bars so the split stays.

Simulation experience ◐ (native model, not tested in game): a soft front and a stiff rear was fastest. Softening the rear bar from the stiff end cost 1–4 s, stiffening the front 0.7–1.2 s. Make a soft front with the bar, not the spring: a stiff front bar with a soft front spring was 1.7 s slower. Balance changes saturate: five changes in the same direction gave −2.3 s one by one but only −1.2 s together.

NOT YET CONFIRMED

- Why a rear-biased split causes braking understeer on a long car is seen in the simulation, but the mechanism is open ◐.
- The bars' effect over bumps and gutters: the flat simulation circle does not show it, so the game decides.

CHAPTER 05

DAMPERS

IN SHORT

In the native model the dampers are a small lever for lap time but a big one for feel. In our experience stock, current and maximum damping were within half a second of each other, while minimum damping made the car unstable or undrivable ◐. Springs and bars decide how much load finally moves, the dampers decide how fast. That is why dampers act in transitions (braking, turn-in, opening the throttle, bumps), not in the middle of a long corner.

The game's knee is high (0.1 m/s), so the slow values do almost all of the handling work and the fast values only bumps, kerbs and gutters.

> Markers: ● confirmed in game or in the game code, ◐ native simulation or inferred, not tested in game, ○ real-car background.

HOW THE GAME CALCULATES IT

1. Damper force depends on the wheel's speed v, not its position. Below 0.1 m/s: F = slow · v. Above 0.1 m/s: F = slow · 0.1 + fast · (v − 0.1) ●. Fast is the slope of the curve after the knee, not a separate force.
2. Bump (compression) acts when the wheel rises towards the body. Rebound acts when the wheel drops away from the body ●.
3. The values act directly at the wheel; there is no motion ratio ●. The slider multiplies the value by 1.65 per step, ±3 steps ●. On one car the displayed number is 2 × the native coefficient, but this has not been checked on other cars ◐.
4. The knee is fixed at 0.1 m/s ●. Real dampers typically have their knee at 0.025–0.075 m/s ○, so in the game the slow range also covers much of the road's unevenness.
5. The roll centre is at ground level, so all lateral load transfer goes through springs, bars and dampers ●. The dampers' share in transitions is therefore probably larger than in a real car ◐.

6. The bump stop is rigid and not adjustable. Travel = spring length − sag ●, and the damper affects how close to the stop the suspension drifts over bumps.

What each range handles:

WHEEL SPEED	DRIVING EVENT	WHICH VALUE
Below 0.1 m/s	Roll at turn-in (about 0.03 m/s ◐), dive under braking, squat on throttle, gentle waves	slow bump and slow rebound
Above 0.1 m/s	Bumps, kerbs, gutters, driving over the verge	fast bump and fast rebound (slow acts up to the knee)
Middle of a long corner	The motion has stopped	none: the dampers do nothing

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SLOW AND FAST BUMP

What it does. Bump slows the wheel rising towards the body. As long as the motion continues, it adds load to that corner's tyre ● (force from speed). The axle with stiffer compression takes a larger share of the load transfer in transitions and briefly loses grip relative to the other ○.

- **Front slow bump** acts under braking and at turn-in, when the nose dives and the outer front wheel compresses.
- **Rear slow bump** acts when opening the throttle and on exit, when the rear squats.
- **Fast bump** acts only when the wheel hits a bump, kerb or gutter.

Ratio. When fast is lower than slow, the curve flattens after the knee (digressive), as in racing dampers ○. The game also allows fast > slow (progressive) ●. In setups that have worked in game, fast bump is about 0.6–0.7 × slow bump ●.

When you raise:

CHANGE	EXPECT	TOO MUCH	EXPERIENCE
Front slow bump	The nose dives more slowly under braking and the car is calmer at turn-in.	Understeer on corner entry; the nose does not take load.	○ Sources agree on the direction. ● A 10 % change was within noise in game.
Rear slow bump	The rear squats more slowly on throttle.	On exit the rear goes light for a moment: oversteer and weaker traction ○.	◐ Both ×0.7 and ×1.3 were slower than the stock value.
Fast bump	The body is supported over bumps and does not sink as deep.	A bump or kerb throws the car up and the tyre leaves the road for a moment.	● Fast bump ×1.5 felt very good in game. ◐ Native effect only −0.05 s.
All bump down	The wheel follows the road and soaks up bumps.	The body sinks fast and hits the bump stop.	◐ Bump at minimum was about two seconds slower.

SLOW AND FAST REBOUND

What it does. Rebound slows the wheel dropping away from the body, so it holds the body down ●. It affects the neighbouring and diagonal corners more than its own ○.

- **Front slow rebound** acts when opening the throttle, as the nose rises, and on the inner front wheel at turn-in.
- **Rear slow rebound** acts under braking, as the rear rises, and on the inner rear wheel at turn-in.
- **Fast rebound** decides how fast the wheel returns to the road after a bump or kerb.

Ratio. In setups that have worked in game, slow rebound is about 3.6 × slow bump at the front and 4.5 × at the rear ●. Fast rebound is about half of slow rebound at the front and a third at the rear ●. Real cars often use rebound 2–3 × bump ○. When rebound grows far above bump, the suspension packs down towards the bump stop over a series of bumps: the wheel drops in fast but returns slowly ◐. In our experience halving a very high rebound cut the time on the bump stop by almost half ◐.

When you raise:

CHANGE	EXPECT	TOO MUCH	EXPERIENCE
Front slow rebound	The nose rises more slowly on throttle and stays loaded, so throttle understeer drops.	The car feels hard and packs down; the inner front goes light at turn-in.	● Halving front rebound made the car more predictable, but brought slight throttle understeer in fast corners. ◐ The native model showed the opposite direction (conflict).
Rear slow rebound	The rear rises more slowly under braking and helps rotate the car at turn-in.	The inner rear goes light; the rear bounces and steps out over bumps.	◐ Lowering it about 25 % has calmed a bouncing rear. ● +20 % on the rear dampers felt good in game.
Fast rebound	The body stays calm after bumps.	The wheel does not return to the road in time and grip breaks over a series of bumps.	◐ Fast rebound ×0.6 was 0.14–0.18 s slower, so too low was slower.
All rebound down	The wheels return quickly and the car does not pack down.	The body wallows and pitches.	◐ All rebound at minimum: the car did not finish the native run.

WORKING TOGETHER: HOW TO SET THEM

Which damper moves in each phase (flat road ○, same mechanics in the game ●):

PHASE	FRONT	REAR	DOMINANT PAIR
Straight braking	bump	rebound	front bump + rear rebound
Braking into the corner (trail brake)	outer bump, inner rebound	outer bump, inner rebound	outer front bump + inner rear rebound
Turn-in, off the brakes	outer bump, inner rebound	outer bump, inner rebound	inner front rebound + outer rear bump
Mid-corner	—	—	no effect
Exit, throttle	rebound	bump	front rebound + rear bump

PHASE	FRONT	REAR	DOMINANT PAIR
Bump, kerb, gutter	fast bump, then fast rebound	same	the wheel that hits

Setup order:

1. Springs and bars first. They decide where the load transfer settles; the dampers only how fast.
2. Level: change all four by the same factor (one slider step = ×1.65). In the native model stock to maximum was within noise, but minimum was too little ◐.
3. Bump/rebound ratio: if the car feels hard and hits the bump stop over bumps, rebound is too high relative to bump.
4. Front/rear: correct the transition balance by symptom (below).
5. Fast values last, for bumps and gutters.

Things to keep in mind:

- In roll the outer wheel compresses and the inner wheel extends at the same speed. The axle's transient load transfer is then about (bump + rebound) / 2 × speed, and if rebound is larger, the difference (rebound − bump) × speed pulls the body down ◐ derived.
- When you stiffen a spring but not the dampers, the suspension becomes relatively less damped, because critical damping grows with the square root of the spring ○ (same formula in the game). Raise damping by roughly √(new spring / old spring).
- The ratios above (rebound 3.6–4.5 × bump, fast about 0.6 × slow) come from a preset that has worked in game on several cars. They are a starting point, not a searched-out value ◐.
- Real-car rules of thumb, such as a damping ratio of 0.5–0.7, are background only ○. Front rebound that works in game is still clearly above them.

EFFECT ON OTHER SETTINGS

SETTING	HOW IT LINKS TO THE DAMPERS IN THE GAME PHYSICS
Springs	Suitable damping depends on the spring (critical damping ∝ √(spring · mass)). A soft spring with firm slow damping gives quick support in transitions and a soft feel over bumps ○.

SETTING	HOW IT LINKS TO THE DAMPERS IN THE GAME PHYSICS
Spring length	Travel = length − sag ●. Too much rebound packs the car onto the bump stop, and a short spring brings the stop even closer. In our experience a longer spring cut bump-stop time by three quarters, halving rebound by almost half ◐.
Bars	A bar has no damping of its own ●. A stiffer bar on the same dampers makes roll relatively less damped ○.
Differential	Opening the throttle squats the rear (rear bump) while the diff locks. Rear slow bump and throttle lock both act on the rear on exit, so change only one at a time ◐.
Camber	Camber changes with spring travel ●, and the damper slows that change. The effect is small, because camber's grip effect is small ◐.
Tyres	The tyre's vertical compliance is in series with the suspension; no tyre damping of its own has been found in the native code ◐.
Bumps and gutters	The asphalt bump pattern is −0.0035…0 m (repeat 20 m) and ±0.025 m (repeat 200 m) ●. Gutters and driving over the verge are the main events in the fast range on a line that uses them.

SYMPTOMS

Change one value at a time by about 15–20 % (half a slider step) and drive the same corner. Damper symptoms come in transitions: if the symptom is in the steady middle of a corner, look at springs and bars first.

PHASE	SYMPTOM	WHAT HAPPENS IN THE GAME	CHANGE
Braking into the corner	The nose dives quickly and the rear goes light	The front drops and the rear rises too fast	Raise front slow bump and rear slow rebound.
Braking into the corner	The car will not turn while trail braking	The outer front takes too much load in the transition	Lower front slow bump ○.
Turn-in	The rear steps out right at turn-in	The inner rear goes light; the rear takes too much in the transition	Lower rear slow rebound ◐.

PHASE	SYMPTOM	WHAT HAPPENS IN THE GAME	CHANGE
Exit, opening the throttle	The nose pushes on throttle in fast corners	The nose rises quickly and loses load	Raise front slow rebound ●.
Exit, opening the throttle	The rear goes light and steps out as the throttle comes in	The rear compresses slowly while the diff locks	Lower rear slow bump, or first the diff preload.
Bumps and gutters	The car feels hard and bounces over a series of bumps	Rebound does not let the wheel back; the suspension packs onto the bump stop	Lower slow rebound ● or lengthen the spring.
Bumps and gutters	A gutter or kerb throws the car into the air	Fast bump too high	Lower fast bump.
Bumps and gutters	The car wallows and pitches after a bump	Fast rebound too low	Raise fast rebound.

FINAL TIP

Keep the dampers near stock or the touge weapon preset first: in the native model stock and maximum differed little, but too little damping ruined the car. Then refine one phase at a time: rebound for packing over bumps, front slow rebound for throttle understeer, and fast values for gutters. If the car feels hard, look at the bump/rebound ratio and the spring length before the spring.

In our experience rebound is clearly larger than bump in every working setup, and relatively larger at the rear than at the front. A heavier car uses larger numbers with the same ratios.

NOT YET CONFIRMED

- The direction of front slow rebound on throttle is in conflict: the game and the literature say less rebound adds throttle understeer, the native model showed the opposite ◐. This guide follows the game.

- The ratio between the displayed number and the native coefficient (2 × on one car) is unchecked on other cars, so the guide does not compute damping ratios.
- Bumps, gutters and verges have not been run in the native model, so the fast-value advice rests on one game test and general mechanics.
- The phase table directions come from real-car sources (○). The game's damper formula is the same, but not every direction has been run in the native model.

CHAPTER 06

CAMBER AND TOE

IN SHORT

Camber and toe are small levers in this game. In the native model camber changes lateral grip by at most about 3.5 % and costs braking and traction grip 0.5 % per degree ●. In our experience camber −2 vs −5 was within noise on several cars ◐, but a front camber close to zero has cost more than a second ◐. Toe is a slightly bigger lever: rear toe-in calms the rear and rear toe-out costs grip, but on a full lap the difference stays under a second ◐. Refine bars, springs and the diff first, and use camber and toe to finish.

The game has two differences from a real car. The camber you set is not the tyre's actual camber, because the game adds −3…+5° to it depending on spring extension ●. And the native code has a rigid-axle suspension type on which the camber setting does nothing ●. Which of the game's suspension types (Mcferson or Doublewishbone) maps to the rigid axle is not yet known.

> Markers: ● confirmed in game or in the game code, ◐ native simulation or inferred, not tested in game, ○ real-car theory.

CAMBER: TYRE ANGLE

How the game calculates it:

1. Camber to the body = setting − 3° + 8° · x, where x = current spring length / stroke (0 = bottomed, 1 = fully extended) ●. At rest the actual camber is therefore about 2–3° more positive than the setting. A typical front wheel at rest sits at about setting + 2.4°, so −3.5 is about −1.1° at rest ◐ calculated.
2. Camber changes with travel by 8° / stroke, so with a 100 mm stroke it goes 0.8° more negative for every centimetre of compression ●. In a corner the outer wheel compresses and goes more negative, but the body

rolls outwards and takes it more positive. On one test car these almost cancelled: at the limit the outer wheel was at about setting + 3.4° to the ground ◐.

3. In the tyre, camber does not raise the lateral peak. It stiffens the grip build-up: −5° gives 14 % more cornering stiffness, which feels like a sharper initial response ●. Lateral force also gets a small offset (−3° → +0.15°) with no practical effect ●.
4. Longitudinal grip (braking and traction) is multiplied by 1 − 0.005 · |camber|, so it drops 0.5 % per degree ●. At the front a large camber lengthens braking, at the rear it weakens traction.
5. Native suspension type 1 is a rigid axle: the wheels follow the axle's roll, and the camber setting and camber curve are zeroed ●. The game passes the car data's type to the native code as such, but whether Doublewishbone is type 1 is not known ◐. If it is, camber does nothing on cars with Doublewishbone at that end.
6. The adjustment range comes from the control arm parts. The base part gives −4.5° camber, and an adjustable part gives ±3.5°. With Prototype arms the limit is about −21.5° front and −22.5° rear ●.

When you add negative camber:

CHANGE	EXPECT	TOO MUCH	EXPERIENCE
Front camber more negative	A slightly sharper initial response and a few percent more mid-corner grip.	Braking gets longer and turning no longer improves.	◐ Front 0 → −15: lateral grip +1.1 %. ◐ Front −3.5 → −2 was 0.4–0.6 s slower on two cars, −5 within noise.
Rear camber more negative	The rear holds slightly better mid-corner. Rear camber helps grip more than front camber.	The rear wheels spin on exit, because traction drops.	◐ Rear −13: lateral grip +3.4 %. ● Rear −3.3 together with other changes has worked in game.
Camber towards zero	Braking and traction grip improve slightly.	Lateral grip drops and the initial response gets lazy. Positive camber clearly hurts.	● Rear −4.3 → −3.5 together with another change was even in game. ◐ +10° −6.5 % (old run, only the direction is reliable).

TOE: WHEEL DIRECTION

How the game calculates it:

1. Toe is set in degrees. Plus is toe-in (the front edges of the wheels point inwards) and minus is toe-out ●. The game adds toe to the wheel angle, −toe on the left wheel and +toe on the right, front and rear ●.
2. At the front, toe comes on top of the Ackermann angle ●. Ackermann is its own setting (0 = full Ackermann, 100 = parallel wheels), and it had no measurable grip effect in the native model ◐.
3. In a corner, rear toe-in turns the loaded outer wheel slightly into the corner, so the rear makes more lateral force and the car stabilises ○ theory that fits the native result (rear toe-in raised steady cornering grip).
4. On a straight, toe means the wheels roll at a small slip angle in opposite directions. The lateral forces cancel, but a small drag remains ◐ inferred, not measured.

When you add:

CHANGE	EXPECT	TOO MUCH	EXPERIENCE
Rear toe-in (+)	The rear calms down when braking into a corner and in long corners, and cornering grip rises in big corners.	The car understeers and will not rotate in tight corners, and the lap gets slower.	◐ +0.2 cut the sideways slide when braking into corners by 4–8 %, but +0.4 already understeered. ◐ +0.3 / +0.6: grip in fast corners +2…+4 %, but the lap was 0.5–1.1 s slower on most cars.
Rear toe-out (−)	The car rotates more readily.	The rear lets go under braking and in corners, and grip drops.	◐ −0.3: grip in tight corners −3.6 %, in fast corners −3 %. On a full lap anything from neutral to 1.5 s slower, depending on the bars.
Front toe (±)	The effect is small. Toe-out sharpens turn-in ○.	The car wanders on the straight and feels nervous ○.	◐ ±0.3 was 0.2–0.8 s slower in both directions on most cars and faster on one.

EFFECT ON OTHER SETTINGS

Camber depends on the suspension, so set it after springs and bars. Toe is nearly independent of other settings, but rear toe-in does the same job as settings that stabilise the rear.

SETTING	HOW IT LINKS TO CAMBER AND TOE IN THE GAME PHYSICS
Springs (length and rate)	Rest camber depends on sag. A stiffer spring or a longer length makes camber more positive: a much stiffer front spring about +0.8°, +8 mm length about +0.2° ◐ calculated. Check camber after big spring changes.
Bars and roll	The body rolls out and takes the outer wheel towards positive camber. Stiffer roll keeps camber closer to the setting ◐. Balance is still found with the bars, not with camber.
Caster	When steering, the outer wheel gains negative camber of about caster · sin(wheel angle) and the inner wheel positive ●. Caster 8.4 and a 30° wheel angle give about 4°. Caster had no grip effect in the native model ◐.
Ackermann	Front toe is added on top of the Ackermann angle ●, so change them one at a time.
Differential and brakes	Camber costs 0.5 % longitudinal grip per degree ●. Rear camber weakens exit traction in the same direction as a too-high throttle lock, and front camber lengthens braking. Rear toe-in calms trail braking in the same direction as a more front-biased brake balance ◐.
Tyres	The camber coefficient is the same on all tyres ●. In the game formula, width does not change the camber effect ◐.
Suspension type	A rigid axle (native type 1) zeroes camber ●. Check your car's suspension type before refining camber (see Not yet confirmed).

SYMPTOMS

Change camber by 0.5° and toe by 0.1° at a time, and drive the same corner in the same gear. If the symptom responds to bars, springs or the diff, use those first.

PHASE	SYMPTOM	WHAT HAPPENS IN THE GAME	CHANGE
Braking into the corner	The rear wanders or tries to overtake when you brake into the corner	Weight moves forward and the rear wheels' lateral force is not enough	Rear toe +0.1…+0.2 ◐. Try brake balance and decel lock first.
Braking into the corner	The front wheels lock or braking gets long	Large front camber costs longitudinal grip	Front camber 0.5–1° towards zero.
Turn-in	The car is lazy to turn	The front tyre's cornering stiffness is low	Front camber −0.5°. Front toe −0.1 helps in theory ○, not in the native model. Try the bars first.
Mid-corner	The rear slides in a long corner	The rear wheels run out of lateral force	Rear toe +0.1…+0.2 or rear camber −0.5°.
Mid-corner	The car understeers in a long corner even though the bars are set	Rear toe-in holds the rear too much	Rear toe towards zero.
Exit	The rear wheels spin as you straighten and open the throttle	Rear camber costs traction	Rear camber 0.5° towards zero ●.
Straight and top speed	The car wanders or feels nervous	Toe-out at either end ○	Front toe 0, rear toe 0…+0.1.
Bumps, kerbs and gutters	The feel changes as the suspension compresses	Camber changes with travel by 8° / stroke ●	Do not correct with camber; use the suspension.

FINAL TIP

Camber −3.5 / −3.5 has been in every working setup of the touge weapon preset in game ●. On screen it looks like a lot, but at rest it is about 2–3° more positive than the number. Keep front toe at zero and rear toe at 0…+0.1, and change them only by symptom.

If you do not know whether camber affects your car, do a quick test: front camber −3.5 → −8 and drive the same corner. If the feel does not change at all, the front axle is probably the native rigid axle.

Simulation experience ◐ (native model, not tested in game): front camber −4.6 → 0 was 1.4 s slower, mostly in medium-speed corners, and with a stiff front bar even more (2.3 s together). −6.5 brought nothing over −4.6. So front camber is more than a finishing touch when it starts near zero. Rear camber and front toe did not show in lap time. Rear toe-in +0.3 was 0.7 s slower.

NOT YET CONFIRMED

- The suspension type is the main open question: is Doublewishbone native type 1, the rigid axle? The game metadata is encrypted, so a game test decides it (front camber −3.5 vs −8).
- The rest camber examples assume the stroke equals the spring length. That is inferred from the bottoming formula, not read from the native code.
- The grip results are from the native model. There is no separate game test of camber or toe alone; the game tests changed another setting at the same time.
- Toe drag on the straight is inferred, not measured.
- The front toe effect on turn-in is real-car theory ○, and the native model did not support it.

CHAPTER 07

DIFFERENTIAL: THROTTLE LOCK, DECEL LOCK AND PRELOAD

WHAT EACH SETTING DOES

This guide is based on the game's own differential code and on native simulation runs. Real-diff theory is used only where the game works the same way.

How the game's clutch diff calculates it:

1. The driveline torque T is first split evenly, T/2 to each driven wheel ●.
2. The lock then moves torque from the faster-spinning wheel to the slower one, up to the lock torque T_lock ●.
3. On throttle T_lock = preload + throttle lock × T. On engine braking T_lock = preload + decel lock × |T| ●.
4. The wheel torque difference can be 2 × T_lock, so the largest split is (1 + 2L) : (1 − 2L). Lock 0.25 = 3 : 1, 0.30 = 4 : 1, and 0.50 or more = all torque can go to one wheel ●.
5. T is the driveline torque. The pedal brakes do not pass through the diff, and with the clutch pressed no torque is transmitted. Then the only lock is the preload ●.

SETTING	WHAT IT IS IN THE GAME	WHERE IT ACTS	WHAT IT DOES
Throttle lock (Lock on acceleration)	Share of engine torque that locks the wheels	On throttle, most on exit	When the inner wheel starts to spin, torque moves to the outer wheel. More traction and the rear pushes. Too much: the rear steps out at full throttle.

SETTING	WHAT IT IS IN THE GAME	WHERE IT ACTS	WHAT IT DOES
Decel lock (Lock on deceleration)	Share of engine-braking torque that locks the wheels	Off throttle, corner entry	The outer wheel brakes more, so the car stabilises. More = more stable but lazier to turn.
Preload	Constant lock in newton-metres (Ultimate 0–520)	Always, also with the clutch pressed	Does not grow with torque. So it dominates when there is little torque: light throttle, the first opening of the throttle, engine braking and shifting.

The diff part sets the range ●:

DIFF PART	THROTTLE LOCK	DECEL LOCK	PRELOAD	RESOLUTION (LOCK / PRELOAD)
Ultimate (tier 5)	0–1.0	0–1.0	0–520	0.005 / 3.25 Nm
Prototype (tier 6)	0.6–1.6	0.3–1.3	0–100	0.0125 / 1.25 Nm

With the Prototype diff the throttle lock is always at least 0.6, and in the native code a lock of 0.5 or more is a full lock. The throttle side is therefore always fully locked. An adjustable throttle lock (for example 0.24–0.30) needs the Ultimate diff.

Order of magnitude (◐, assumed engine torque 250 Nm): in 2nd gear the axle torque is about 250 × 2.00 × 4.00 = 2000 Nm. Throttle lock 0.3 gives about 600 Nm of lock at full throttle, and preload 100 is only 1/6 of that. Engine-braking torque is much smaller, so decel lock 0.15 gives only tens of newton-metres. Then preload 100 can be larger than the whole decel lock.

When you raise:

CHANGE	EXPECT	TOO MUCH	EXPERIENCE
Throttle lock	On exit the inner wheel spins less, the car pulls better, and the rear starts to push.	The rear steps out at full throttle in a tight hairpin.	◐ In a tight hairpin 0.15 was about 0.4 s quicker than 0.25, and 0.40 spun. Over a full lap 0.10–1.0 was within noise and 0 clearly slowest. ● Values around 0.25–0.30 have worked in game on several cars.
Decel lock	Corner entry stabilises, and the rear calms down under braking and on downshifts.	The nose pushes off throttle, and the lock fights rotation on the brakes.	◐ In a tight hairpin 0–0.8 differed by under 0.1 s; only the rear wheel speed difference changed. ● In our experience a low value suits a style that rotates the car on the brakes.
Preload	Transitions smooth out: the inner wheel spins less and the rear calms down with the clutch pressed.	The rear jerks as the throttle opens, and the car understeers on light throttle. A tight hairpin spun at 150 or more.	◐ 0–520 was within noise over a lap; the rear wheel speed difference dropped from 0.58 to 0.27. ◐ Lowering preload and lock slowed the growth of a slide (55 → 42–44 °/s). ● Preload 0 has also felt good in game.

SETUP ORDER AND WHAT ELSE MATTERS

Set the throttle lock first, the decel lock second and the preload last. Each has its own phase, and the preload refines the transitions between them.

1. **Throttle lock** decides the exit. In the native model 0 is clearly slowest; in a tight hairpin lower values (about 0.15) were quicker, in a medium corner about 0.25. Over a lap 0.10–1.0 is within noise.
2. **Decel lock** decides the entry. In a tight hairpin 0–0.8 differed by under 0.1 s. Corner angle and rotation stayed the same; only the rear wheel speed difference changed. When pumping the brakes, the lock calms the rear.
3. **Preload** refines the transitions. In the native model 0–520 was within noise for time, but the rear wheel speed difference on throttle dropped

from 0.58 to 0.27. Inner-wheel spin fell from about 45 % to 23 %. In a tight hairpin 150 or more spun at full throttle.

What else affects how the diff feels:

SETTING	HOW IT LINKS TO THE DIFF IN THE GAME PHYSICS
Gear ratio and gear	The lock is a share of axle torque. A low gear multiplies torque, so the same throttle lock locks much more in 2nd than in 4th. Tight hairpins are driven in a low gear, so that is where the lock is felt most. Preload does not change with gear.
Throttle technique	With on/off throttle the torque jumps straight to full, so the throttle lock decides the behaviour and the preload shows only at the start of the opening. On part throttle the preload weighs more.
Clutch and shifting	With the clutch pressed T is zero, so the only lock is the preload. On clutch shifts and downshifts under braking the preload is the only lock.
Brake balance	The brakes do not pass through the diff, but a rear-biased balance slows the rear wheels. In the native model front 40 gave 10–15 % more rotation and 1–2° more entry angle than 60. A rear-biased brake and a low decel lock rotate the car most together.
ABS	ABS limits rear wheel locking, so extreme brake balances did not change lap time in the native model.
Rear bar and rear springs	A stiff rear unloads the inner wheel, so it spins more easily and the throttle lock engages earlier. A softer rear needs less lock.
Engine torque	More torque = more throttle lock in newton-metres at the same setting. Preload is fixed, so on a powerful car its relative share shrinks.

SYMPTOMS

First find the phase where the symptom appears. The phase tells you which lock is active. Change one setting at a time: throttle and decel lock by 0.05, preload by about 50.

PHASE	SYMPTOM	WHAT HAPPENS IN THE GAME	CHANGE
Corner entry, off throttle	The nose pushes and the car will not turn	The outer wheel spins faster and the lock brakes it, making a straightening moment	Lower decel lock. If it is already low, lower preload.

PHASE	SYMPTOM	WHAT HAPPENS IN THE GAME	CHANGE
Corner entry, off throttle	The rear moves or kicks under braking and on downshifts	The rear wheels turn at different rates and the lock does not even them out	Raise decel lock or preload. Move brake balance forward.
Clutch pressed (shift, braking)	The rear is restless only with the clutch pressed	The only lock is the preload	Raise preload.
Mid-corner, light throttle	The car understeers and wants to go straight	The outer wheel is faster, so the lock moves torque to the inner wheel. Here the lock is mostly preload.	Lower preload.
Opening the throttle	The inner wheel spins, revs rise but the car does not pull	Not enough lock to move torque to the outer wheel	Raise preload (start of the opening) or throttle lock (full throttle).
Opening the throttle	The rear jerks as soon as the throttle comes in	Preload ties the wheels together before torque rises	Lower preload.
Exit, full throttle	The rear steps out and the car spins in a tight hairpin	The lock sends almost all torque to the outer wheel	Lower throttle lock or preload.
Exit, full throttle	The car does not pull out and the inner wheel spins in a wider corner	The lock does not even out the wheels in time	Raise throttle lock by 0.05.

FINAL TIP

In the native model the diff settings hardly changed lap time; only a throttle lock of 0 was clearly slow. Choose the values by what is easiest to control. A steady drift was slower than grip in the native model (−15 % in a tight corner, −6 % in a medium one), but about 20° of slide on exit at full throttle was quickest.

- Always test in the same corner in the same gear, because the gear sets the lock force.

- If the car spins at full throttle, look at the throttle lock first and then the preload. Both push the rear out.
- In our experience a throttle lock of 0.25–0.30, a decel lock of 0.12–0.20 and a preload of 45–100 has been an easy-to-drive starting range in game ●. Touge's diff presets cover this range and the softer and stiffer ones around it.

Simulation experience ◐ (native model, not tested in game): once the throttle lock is above 0, the locks and preload are feel settings: throttle lock 0.15–1.0, decel lock 0–0.5 and preload 0–250 all stayed within 0.4 s. An open diff (throttle lock 0) was 2.4 s slower, mostly in slow corners. The exception is a stiff rear bar, which wanted a throttle lock of at least about 0.3 (see the anti-roll bar guide).

NOT YET CONFIRMED

- The torque T is inferred to be the whole axle torque, not verified.
- The native driver handles a slide differently from a person, so the feel has to be confirmed in game.

CHAPTER 08

BODY DIMENSIONS AS A SETUP AID

IN SHORT

You can hardly adjust the body dimensions, but they tell you where to start your setup. The biggest lesson is the wheelbase: a rear-biased bar split from a short roadster did not carry over to longer cars, and a split corrected for wheelbase has worked in game on nine cars ●. Track width and centre of gravity height decide how much load moves, and mass decides how stiff the springs must be for the same feel.

The only adjustable dimensions are track width (fender part and spacers), centre of gravity (body Stage, weight reduction and spring length) and mass. Everything else comes with the choice of car.

> Markers: ● confirmed in game or in the game code, ◐ native simulation or inferred, not tested in game, ○ real-car theory.

WHEELBASE

How the game calculates it:

1. Longitudinal load transfer is m · a · h / L ●. A short car moves more weight forward under braking and unloads the rear more than a long one.
2. A short car rotates more easily, because the tyres' lever arm around the car's vertical axis is shorter ○. That is why a short car turns with only a little rear movement, and a long car needs more front.
3. Wheelbase is part of the Ackermann geometry (wheel angle depends on L and track) ●, but Ackermann had no grip effect ◐.
4. In our experience the front share of roll stiffness (LLTD) minus the front weight share moves forward as the wheelbase grows ●. A short car has worked with a rear-biased split, a long one with an even or front-biased split.

What has worked in game, by wheelbase:

WHEELBASE	FRONT SHARE OF ROLL STIFFNESS − FRONT WEIGHT SHARE
about 2.3 m	about −0.13 (clearly rear-biased)
2.4–2.5 m	about −0.09…−0.06
2.5–2.6 m	about −0.05…−0.02
2.7 m	about +0.02…+0.03 (front-biased)

Bar numbers do not compare from one car to another, because springs, track, mass and centre of gravity also affect them. Compare the split, not the number.

In the gearing, short cars gained most from an about 10 % longer final drive ◐. This cannot be separated from the rev limiter, because the short cars also had the highest limiter.

TRACK WIDTH AND OFFSET

How the game calculates it:

1. Lateral load transfer is m · a · h / t ●. A wider track moves less load to the outer wheel, and since the tyre loses friction as load grows, grip rises.
2. Track = the car's base width + offset, and offset = the body kit's fender part + spacer (cm) ●. The body kit changes nothing else in the physics, for example not drag ●. The white offset number in Part Setup is rounded to an integer ●.
3. Tyre width does not move the wheel's mounting point ●. The offset gain shrinks with a wide tyre: +0.20 m on both axles gave +2.6 % on 185 mm, +1.9 % on 245 mm and +1.6 % on 285 mm ◐.
4. A wider axle is stiffer in roll (spring and bar stiffness × t^2), so it takes a larger share of the roll moment ●. Its grip still rises: rear +0.20 m moved the front share 0.555 → 0.520 and added 2.7 % grip, front +0.20 m only 0.5 % ◐.
5. Rule: tyres must not stick out of the fenders. In practice this limits the offset.

When you widen:

CHANGE	EXPECT	TOO MUCH	EXPERIENCE
Both axles	Grip rises slightly in all corners.	Tyre outside the fender (against the rule).	◐ +0.10 m +1.4 %, +0.26 m +3.2 %. ◐ On a full run the gain showed only with narrow tyres; otherwise it was within noise.
Rear only	The rear holds better and the exit calms down.	The car starts to understeer if the nose is already at its limit.	◐ Rear +0.20 m +2.7 %.
Front only	Small grip effect; the nose takes share from the rear.	The nose is heavier to turn.	◐ Front +0.20 m +0.5 %. ◐ On another car a wide track cost front grip: −4 % in tight corners and +3 % in fast ones.

The results differ between cars, so the offset gain is not certain on every car ◐.

CENTRE OF GRAVITY HEIGHT

How the game calculates it:

1. Centre of mass = body box floor + mass_up_percent × box height ●. "Center of Mass Height" in Part Setup is this percentage, not inches ●. The game shows it mass-weighted, because the engine and gearbox stay at the body's stock value ◐.
2. The centre of gravity changes with the body Stage, weight reduction (×0.99 / 0.97 / 0.95), body reinforcement (×1.007–1.015) and the weight reduction's own slider (step ×1.05) ●. Spring length raises the body and the centre of gravity 1 mm per mm ●.
3. Load transfer is directly proportional to the height h, both laterally (m · a · h / t) and longitudinally (m · a · h / L) ●. Lower is always better, but the range is small: the whole weight-reduction slider moves it about 2.8 cm, and the lap time did not stand out from noise ◐.
4. A tall car needs more roll stiffness for the same roll, a low car less. The "touge weapon (real CoG)" preset scales roll stiffness by the car's centre of gravity: a tall car gets about 30 % stiffer bars, low cars 17–21 % softer ◐ (not tested in game). The older touge weapon uses the same reference height for all cars, and the setups that worked in game were made with it.

5. Aero has no ride-height dependence ●, so a low body does not add downforce.

When you lower the centre of gravity:

CHANGE	EXPECT	TOO MUCH	EXPERIENCE
Weight reduction or slider down	Slightly less roll and load transfer.	No physics penalty. The slider also shifts weight distribution by ±1.6 points.	◐ The whole slider range: lap within noise.
Spring length down	Lower centre of gravity and less transfer.	The suspension hits the bump stop, which costs far more than the centre of gravity gives.	◐ +20 mm length = transfer +16 %, but grip in fast corners did not change. ● A slightly longer front spring has been more than two seconds quicker in game.

So take the centre of gravity into account in the total bar stiffness, but do not chase it at the cost of travel.

MASS AND WEIGHT DISTRIBUTION

How the game calculates it:

1. The tyre loses friction coefficient as load grows, so a heavier car grips relatively less: about −2 % grip per 100 kg ●. Full Prototype weight reduction multiplies body mass by 0.805 ●, and the fuel tank adds 1–4 % to body mass ●.
2. The spring rate is the wheel rate directly, so a heavier car needs a stiffer spring for the same feel ●. Natural frequency $f = \sqrt{(k / m)} / 2\pi$, where m is the mass on one wheel ◐ calculated. The spring needed for the same frequency grows directly with wheel mass.
3. Suitable damping grows with $\sqrt{(k \cdot m)}$ ○. Touge moves dampers to another car by damping ratio ζ, so a heavier car automatically gets larger numbers ◐.
4. Weight distribution (front weight) splits the static load. The wheelbase relation (front share of roll stiffness − front weight) is relative to front weight, so a nose-heavy car needs a more front-biased bar split for the same balance ●. The weight-reduction slider moves the distribution by ±1.6 points ●.

What it means:

FACTOR	WHAT TO EXPECT	EXPERIENCE
Heavier car	Stiffer springs and larger dampers for the same feel, and slightly less grip.	● A car about 25 % heavier has worked with springs about 35 % stiffer.
Front weight up	The rear calms down on throttle and the nose pushes more easily.	◐ A nose-heavier car did not spin at full throttle in a tight corner where a 50/50 car did. ◐ More front weight was two seconds quicker with stock bars, no difference with a corrected bar split.
Lighter car	More grip relative to mass and shorter braking.	● Formula −2 % per 100 kg.

The weight distribution result shows that the bar split and the weight distribution do the same job. Once the split is right, a change in distribution no longer shows ◐.

FROM DIMENSIONS TO STARTING VALUES

This is how to move a working setup to a new car using its dimensions. The touge weapon preset does steps 2–6 automatically, and setups made this way have worked in game ●.

1. **Note the car's data:** wheelbase, front and rear track, mass, front weight and Center of Mass Height (Part Setup).
2. **Springs from mass:** keep the same natural frequency as on the working car. A wheel about 30 % heavier needs a spring about 30 % stiffer ◐ calculated.
3. **Spring length from the car's default:** front ×1.07 and rear ×0.90, as in touge weapon ●.
4. **Bar split from wheelbase:** use the table in the Wheelbase section. The total stiffness comes from the roll you want, and a high centre of gravity needs more ◐.
5. **Track inside the fender:** offset as wide as the fender allows. On a rear-drive car widening the rear track helps more than the front ◐.
6. **Dampers by damping ratio:** the same ζ as on the working car, so the numbers grow with √(k · m) ◐.

7. **Gearing:** on a short car, try an about 10 % lower final drive (FD × 0.90) ◐.

After this, refine by symptom: bars first, then the diff, then springs and dampers.

SYMPTOMS

These symptoms come from a setup made for a car of different size. Correct them with bars, springs or the diff, because the dimensions can hardly be changed.

PHASE	SYMPTOM	WHAT HAPPENS IN THE GAME	CHANGE
Braking into the corner	A long car understeers on the brakes and has to be forced to turn	A short car's rear-biased bar split does not rotate a long car	Move the bar split forward by wheelbase ●.
Mid-corner, high speed	Long car: the rear lets go suddenly in a big corner	The limit comes from the rear when the split is too rear-biased	Same fix: split forward ●.
Exit	A short car with a light rear spins on throttle	Low front weight and a short wheelbase make the rear sensitive	Lower diff preload and throttle lock ◐, or widen the rear track.
Braking	A short car dives and the rear goes light	Longitudinal transfer m · a · h / L is larger on a short car	Move brake balance rearwards and add front spring length.
Turn-in	A tall car rolls and is lazy	A high centre of gravity adds lateral transfer	Raise total bar stiffness with the same split.
Bumps and kerbs	A heavy car bottoms out and wallows	Springs and dampers are too soft for its mass	Springs by frequency, dampers in the same ratio.

FINAL TIP

On a new car, look at the wheelbase first. It tells you how far to move the bar split of a working setup, and in our experience this alone cured a nervous rear and braking understeer on every car of the Touge weapon set ●. Mass tells you the level of springs and dampers, and the centre of gravity the total bar stiffness.

In our experience every 10 cm of extra wheelbase has moved the working roll-stiffness split about 4 points forward, up to about 5 points ahead of the front weight share ●.

NOT YET CONFIRMED

- The wheelbase relation was fitted on three cars and tested in game on nine. Cars over 2.7 m have not been tried, so the upper end is an estimate.
- The frequency and damping ratio examples are calculated. Unsprung mass has not been subtracted from the wheel mass.
- The offset gain was positive on one car and negative on another, and the reason is open.
- The corrected centre of gravity calculation (touge weapon, real CoG) is untested in game.
- The link between a short wheelbase and longer gearing may come from the rev limiter.

CHAPTER 09

GLOSSARY

SETTINGS: THE GUIDE'S WORDS AND THE GAME'S NAMES

The guide uses short everyday words. This table shows the name of each setting in the game's own menus, as Touge also shows them in the To game view.

IN THIS GUIDE	NAME IN THE GAME	WHAT IT IS
Anti-roll bar, bar	Anti-roll bar: Stiffness	A torsion spring between the left and right wheel of one axle. Game value × 2000 = N/m.
Spring length	Spring Length	Ride height and suspension travel in millimetres.
Spring, spring rate	Spring Stiffness	Wheel rate. Game value × 1000 = N/m.
Slow bump, slow rebound	Slow Bump, Slow Rebound	Damping below the 0.1 m/s knee: body motion in braking, turn-in and on throttle.
Fast bump, fast rebound	Fast Bump, Fast Rebound	Damping above the knee: bumps, kerbs and gutters.
Camber	Camber	Tyre angle; the game adds −3…+5° depending on spring extension.
Toe	Wheel Toe	Wheel direction in degrees; + = toe-in.
Caster	Caster Angle	Steering axis tilt; changes camber when steering.
Ackermann	Ackermann Angle	0 = full Ackermann, 100 = parallel wheels.
Throttle lock	Lock on acceleration	Share of driveline torque that locks the diff on throttle.
Decel lock	Lock on deceleration	Share of engine-braking torque that locks the diff off throttle.

IN THIS GUIDE	NAME IN THE GAME	WHAT IT IS
Preload	Preload	Constant diff lock in newton-metres, also with the clutch pressed.
Brake balance	Front Brake Balance	Share of brake torque on the front axle.
Offset, spacer	Front / Rear Wheel Offset	Wheel position outwards; track = base track + offset.
Tyre pressure	Pressure	Grip peaks at 2.0 bar; higher pressure lowers rolling resistance.
Tyre profile	Tire profile	Sidewall height in percentage points (±10).
Final drive, FD	Final drive	The last ratio; total ratio = gear × final drive × tyre radius.
Centre of gravity	Center of Mass Height (Part Setup)	A percentage of the body box height, not inches.
Front weight	Weight Distribution in Front	Share of the car's weight on the front axle.

DRIVING PHASES

PHASE	WHAT IS HAPPENING
Braking into the corner	Braking in a straight line and trail braking towards the apex. Weight moves to the front.
Turn-in	Off the brakes and turning in. Weight moves to the outer wheels.
Mid-corner	A steady corner; the body has stopped moving. Springs and bars decide the balance.
Exit	Opening the throttle and straightening. Weight moves to the rear; the diff works.
Straight	Acceleration and top speed. Gearing and drag decide.
Bumps, kerbs and gutters	Fast suspension movements, verges and gutters. Travel and fast damping decide.

TERMS

TERM	MEANING
Understeer	The front gives up first; the car goes wider than you steer. "The nose pushes."
Oversteer	The rear gives up first; the car rotates more than you steer. "The rear steps out."
Load transfer	Weight moving between wheels under braking, acceleration and cornering: m · a · h / track (or wheelbase).
Roll stiffness	How strongly an axle resists body roll: bars + springs, both × track².
Bar split, LLTD	The front axle's share of roll stiffness. The axle with the larger share carries more of the cornering load transfer and lets go first.
Travel	How far the suspension can still compress before the bump stop: spring length − sag.
Bump stop	The rigid end of travel. Hitting it throws the car and costs grip.
Sag	How much the spring is compressed by the car's weight at rest.
Natural frequency	√(spring / mass) / 2π. Keeping it equal moves a spring setup to a heavier or lighter car.
Damping ratio, ζ	Damping relative to the spring and mass; Touge uses it to move dampers between cars.
Knee	The wheel speed (0.1 m/s) where damping changes from the slow to the fast value.
Total ratio	Gear × final drive × tyre radius; the only gearing number the physics uses.
Rim part	The wheel part chosen for a tyre size; it sets the tyre's friction bonus (+0.03…+0.08).
Native model	The game's physics engine run outside the game with a computer driver, used to compare settings evenly.
Within noise	A difference smaller than the normal run-to-run spread, about 0.8 s on a six-minute run.

MARKERS

MARKER	MEANING
●	Confirmed in game or in the game code.
◐	Native model or inferred, not tested in game.
○	Real-car theory, used only as background.

Project Touge is an unofficial fan-made tool for CarX Street. It is not a CarX Technologies product.